

Executive Summary

Titanic Survival Prediction Using Machine Learning

Student Details: Name: Matteddula Mayuri

College: Sri Venkateswara College of Engineering

Task Number: Task 3 – Titanic Survival Prediction

Project Overview:

The Titanic Survival Prediction project aims to predict whether a passenger survived the Titanic disaster using machine learning techniques. The project focuses on data preprocessing, feature engineering, model training, and performance evaluation.

Dataset Description:

The dataset contains passenger information such as passenger class, age, gender, fare, family details, cabin information, and embarkation details. The dataset was analyzed to identify missing values and understand the relationship between various features and survival.

Data Preprocessing:

Missing values in the Age, Fare, and Embarked columns were handled using appropriate statistical methods. Categorical variables were converted into numerical form using Label Encoding to make them suitable for machine learning algorithms.

Feature Engineering:

Additional features were created to improve model performance:

- **Title** extracted from passenger names.
- **FamilySize** calculated using family-related attributes.
- **CabinPresent** indicating whether cabin information is available.

These features helped the model better understand passenger characteristics.

Model Development:

A Random Forest Classifier was used to predict passenger survival. The dataset was divided into training and testing sets, and the model was trained using important passenger attributes.

Model Evaluation:

The model performance was evaluated using accuracy score, classification report, confusion matrix, and feature importance analysis. The results demonstrated the effectiveness of the machine learning model in predicting survival outcomes.

Conclusion:

This project successfully demonstrates the complete machine learning workflow, including data preprocessing, feature engineering, model training, evaluation, and model saving. The developed model can be used for predicting passenger survival and serves as a practical application of supervised machine learning techniques.

Tools Used:

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Google Colab

Author:

Matteddula Mayuri

Artificial Intelligence and Machine Learning